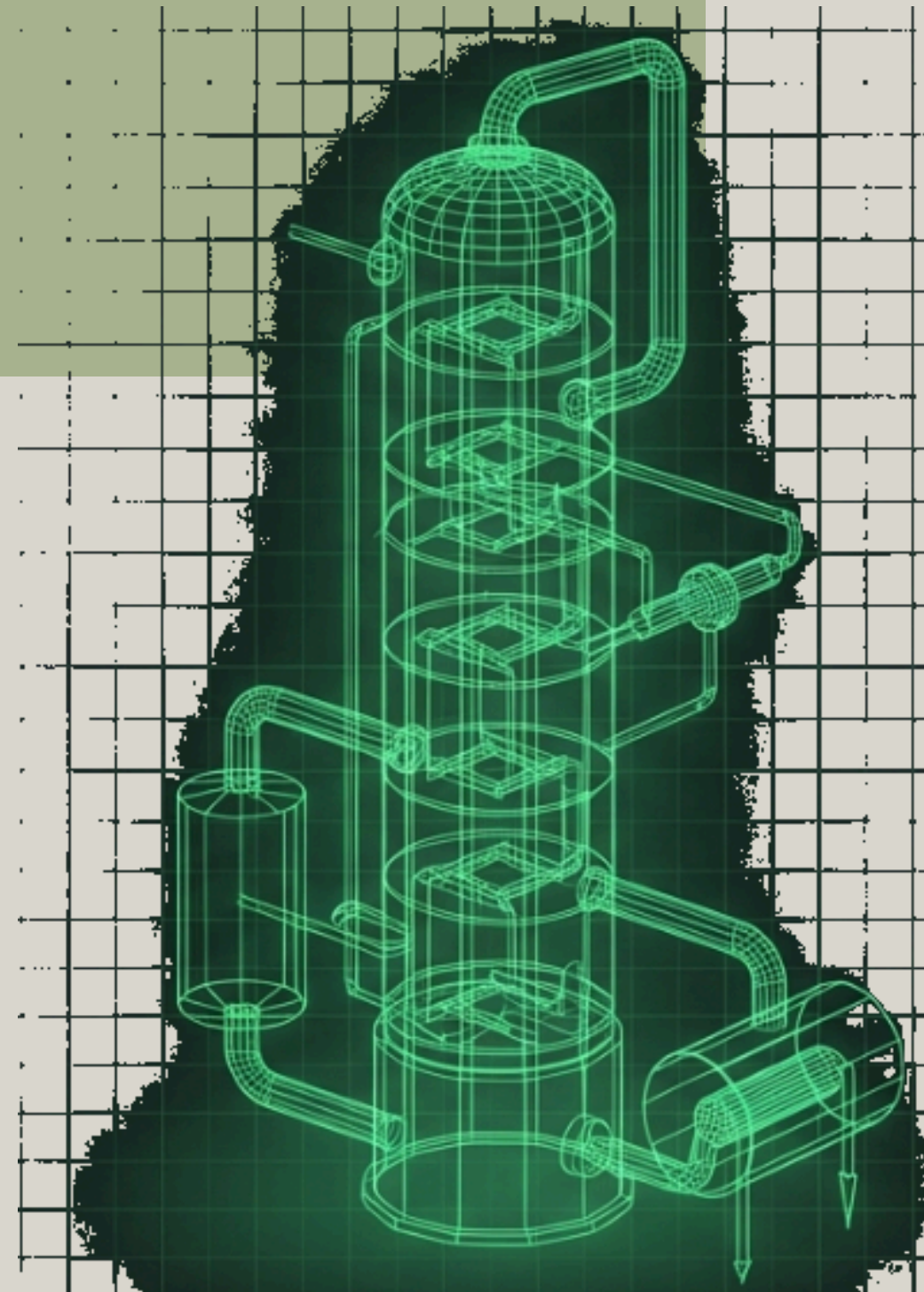


PINCH ANALYSIS TO IMPROVE THE HEAT INTEGRATION EFFICIENCY IN A CRUDE DISTILLATION UNIT

Group 5

- Anuj Kothari- 230008010
- Butle Vedika Ashirwad- 230008012
- Ved pathak- 230008037
- Yashasvi Shukla-2300080040

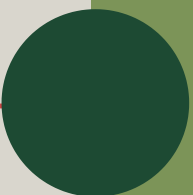




OBJECTIVES



Develop a high-fidelity simulation of a **refinery flowsheet**.



Perform Pinch Analysis to identify minimum utility requirements and maximize internal heat recovery.

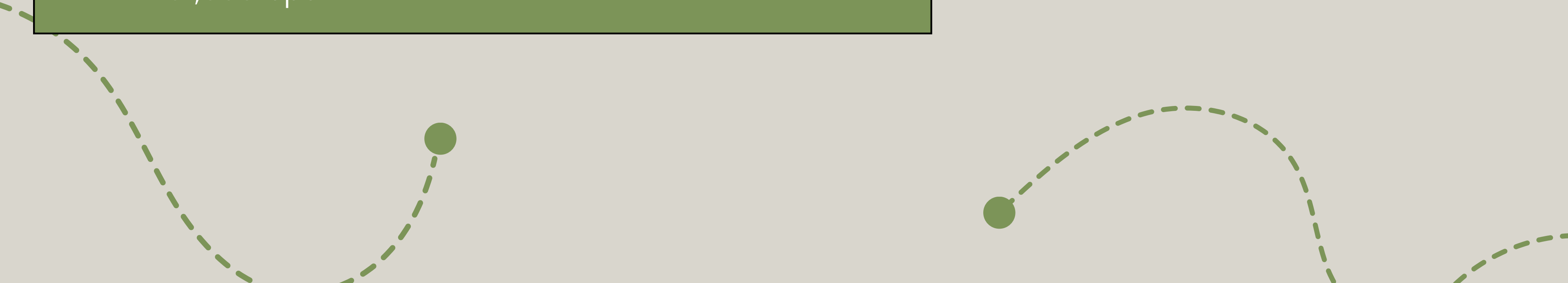


Conduct sensitivity studies to understand the impact of variables like **furnace temperature** on product yields

SIMULATION BASIS & METHODOLOGY

- Software: **Aspen Plus V14**, utilizing the PetroFrac column model.
- Property Method: **Peng-Robinson (PENG-ROB)**, suitable for high-pressure and high-temperature petroleum applications.
- Components: Integrated 12 pure chemical species (Water, Methane through n-Pentane) with narrow-boiling fraction pseudo-components to represent the heavy oil

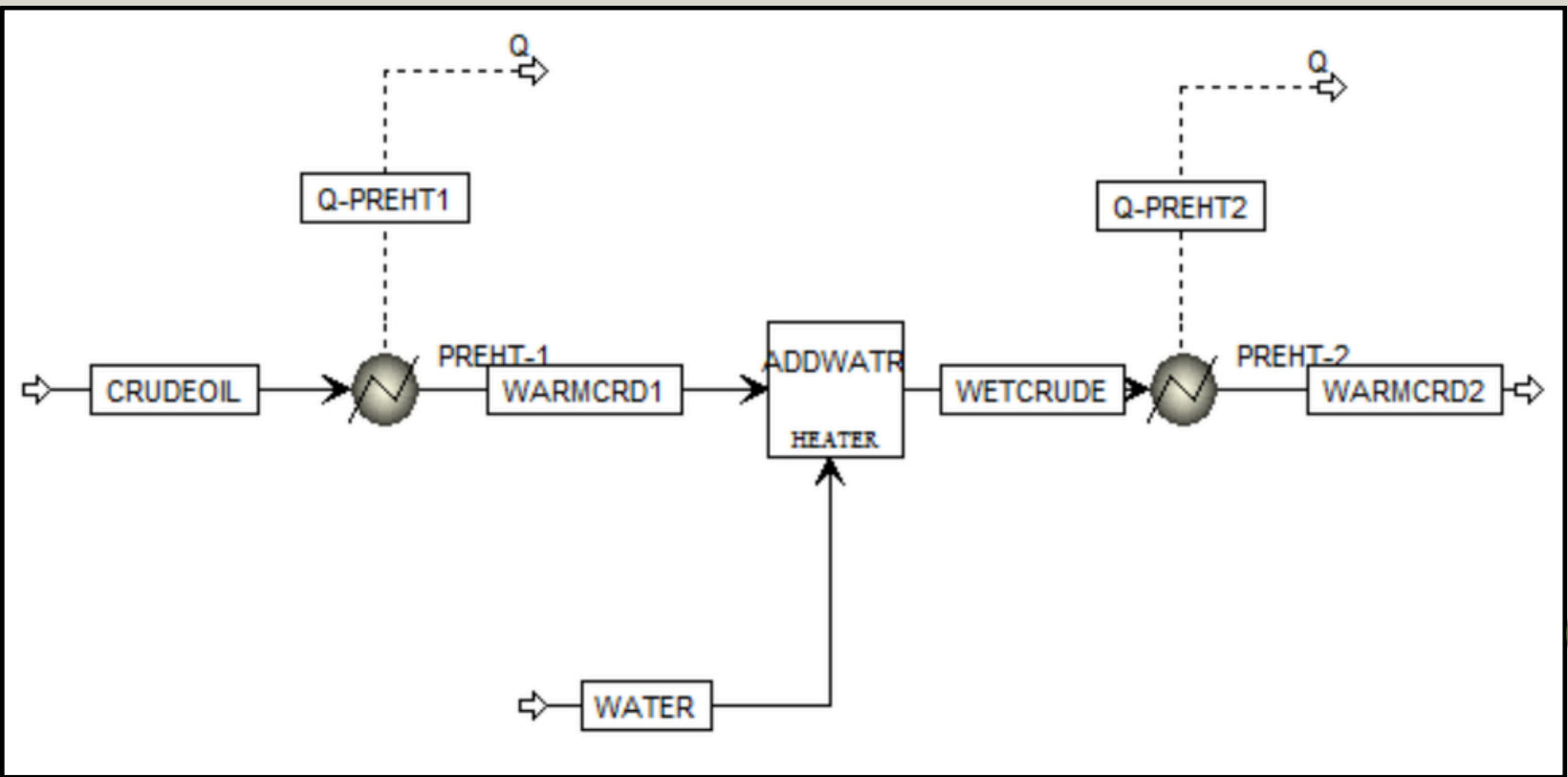
CRUDE CHARACTERIZATION & BLENDING

- **Feedstock:** A blend of Light, Medium, and Heavy crudes
 - **Assay Management:** Defined property curves for each crude, including specific gravity, API gravity, and Sulfur distribution
 - **MIXOIL Blend:** The unit processes a combined stream (MIXOIL) at a standard volume fraction of 1, equivalent to 101,000 bpd
- 

Flowsheet Architecture

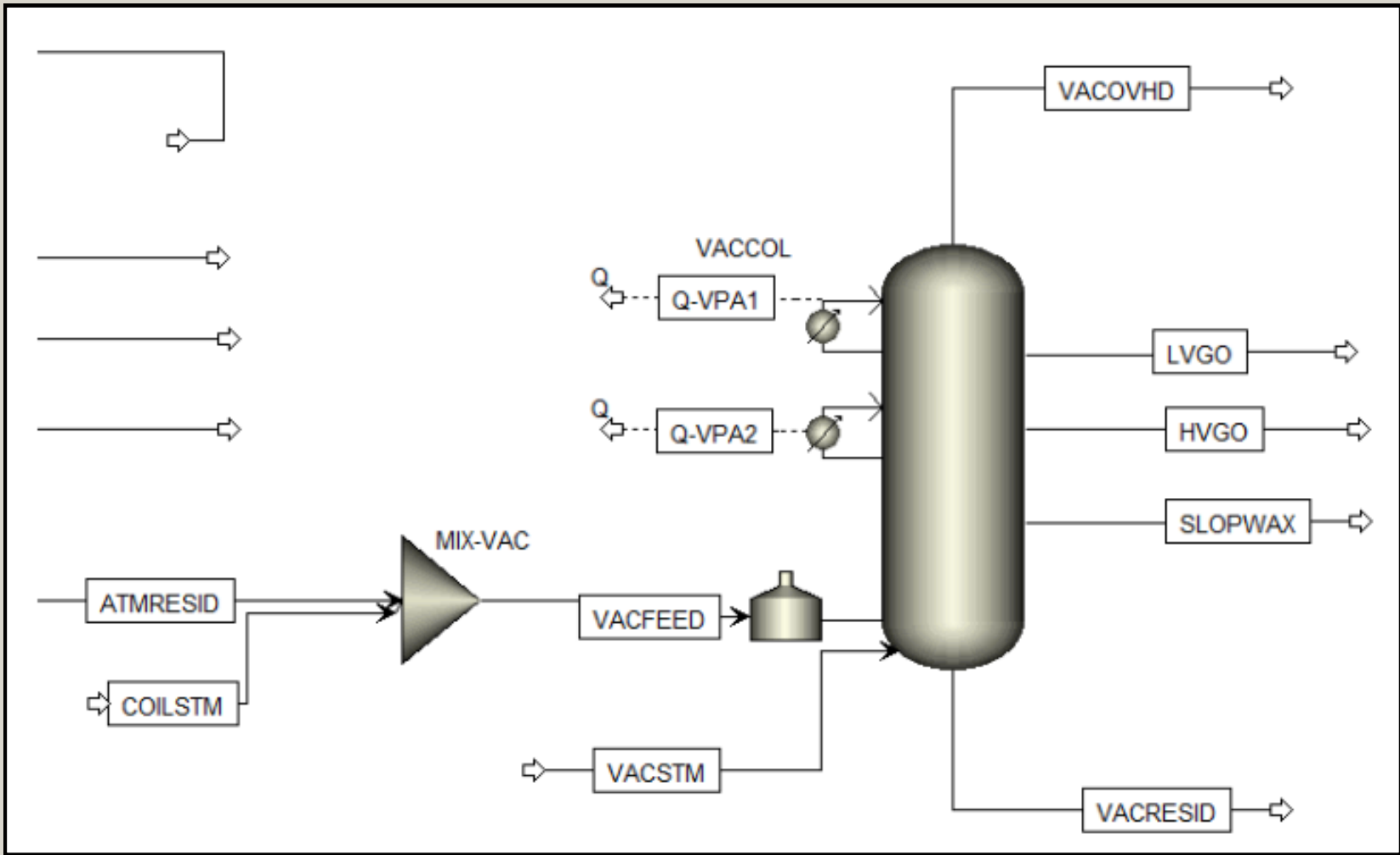
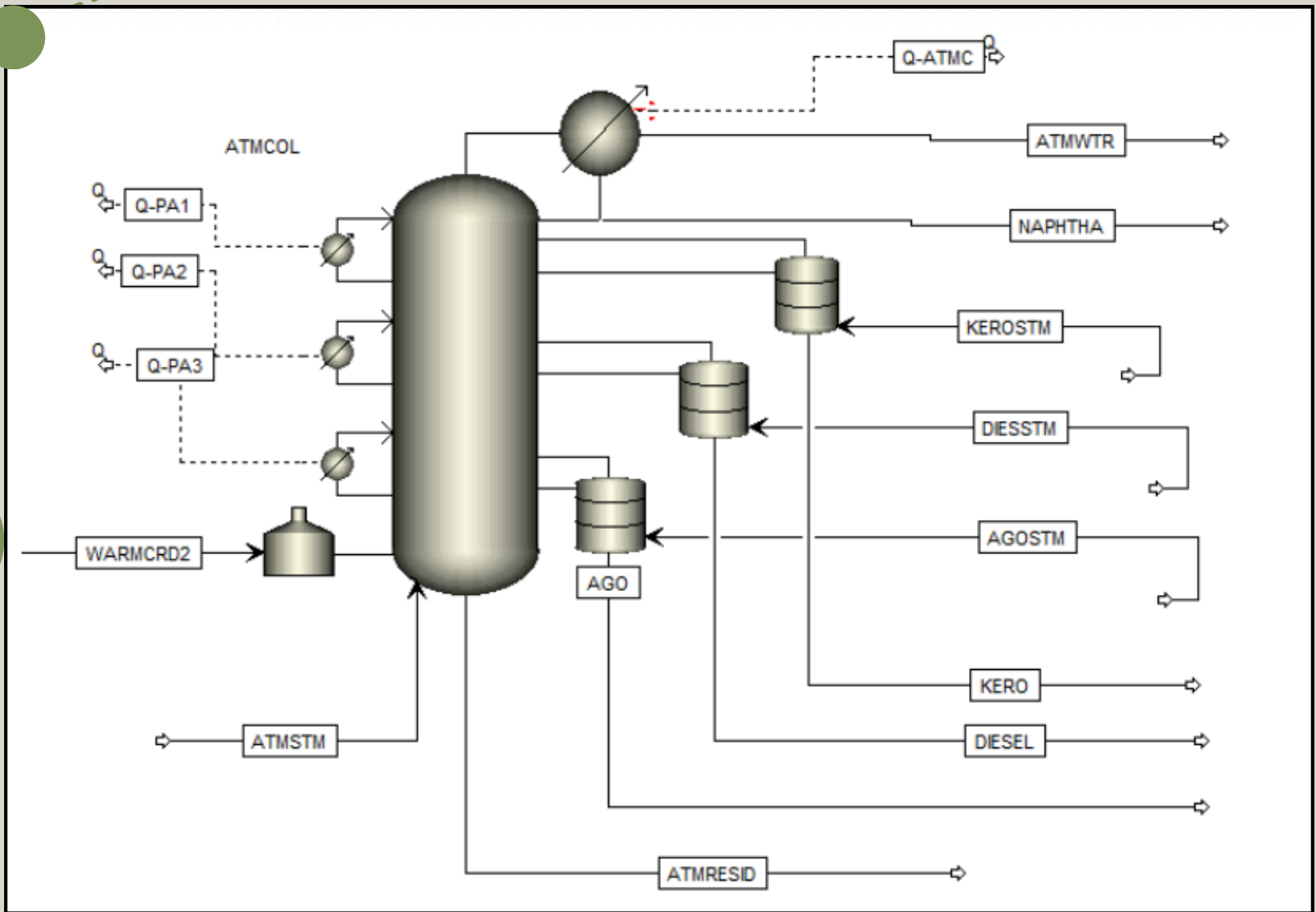
[Input data](#)

Unit Operation	Configuration / Stages	Primary Function	Key Operating Parameters
Preheat Train	Series of exchangers and a Desalter	Heats crude feedstock to required column entry conditions while removing salts and sediments.	Outlet temperature: 450°F; Pressure: 260 psig; includes water injection for desalting.
Atmospheric Column (ATMCOL)	51 total stages (50 trays + 1 condenser)	Primary fractionation of crude into Naphtha, Kerosene, Diesel, and AGO.	Top stage: 250°F (12 psig); Bottom stage: 650°F (22 psig); includes 3 side-strippers and 3 pumparounds.
Debutanizer (DEC4)	47 total stages (45 trays + condenser + reboiler)	Recovers LPGs and stabilizes the overhead Naphtha stream.	Pressure: 150–160 psig; Reflux ratio: 1.5; Feed preheated to 250°F.
Vacuum Column (VACCOL)	14 trays (modeled as non-equilibrium stages)	Recovers heavier gas oils (LVGO/HVGO) from atmospheric residue under vacuum to prevent thermal cracking.	Top pressure: 50 mmHg; Bottom pressure: 62 mmHg; Feed heater outlet: 760°F



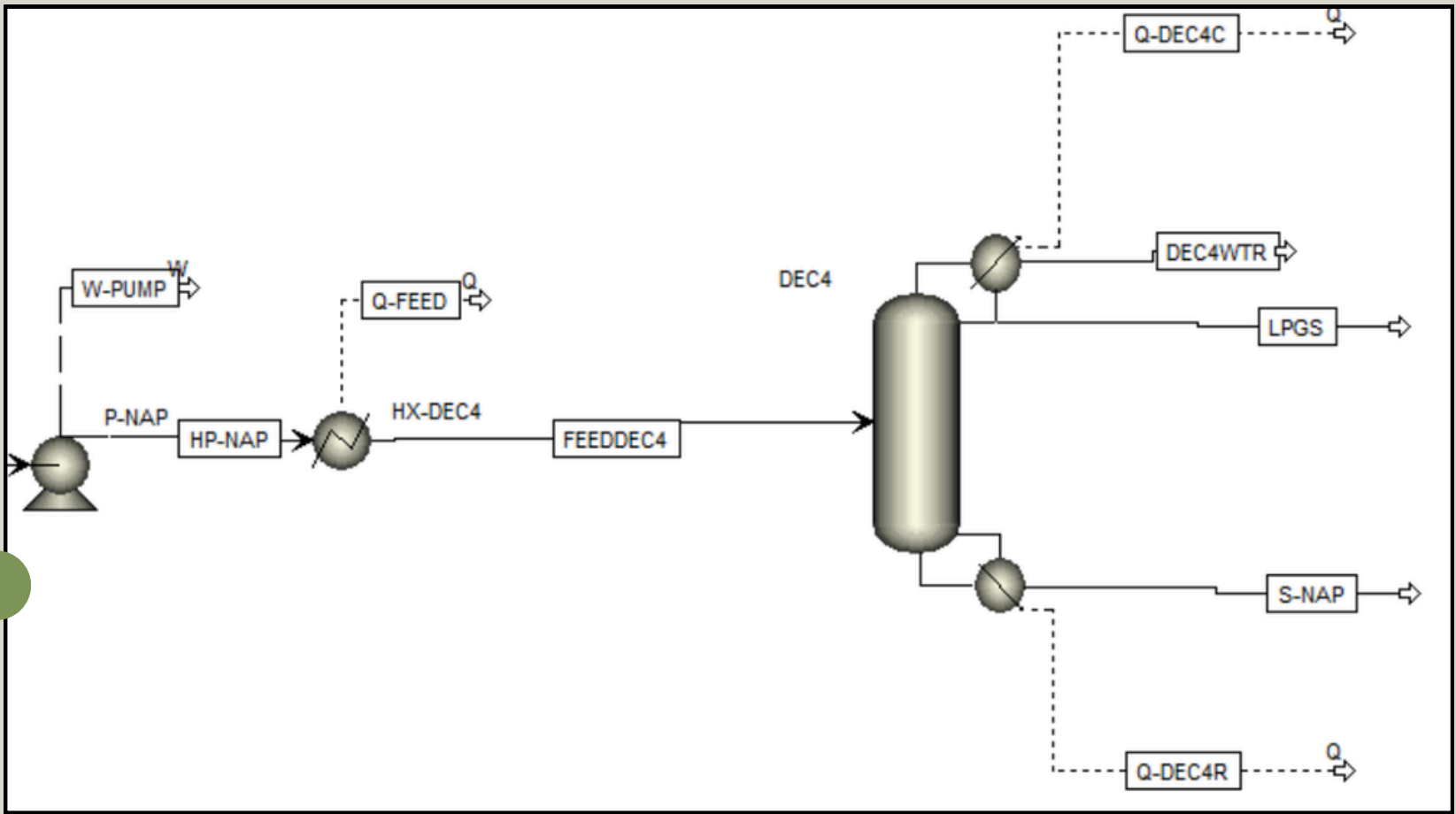
Preheat Train

Atmospheric Column (ATMCOL)

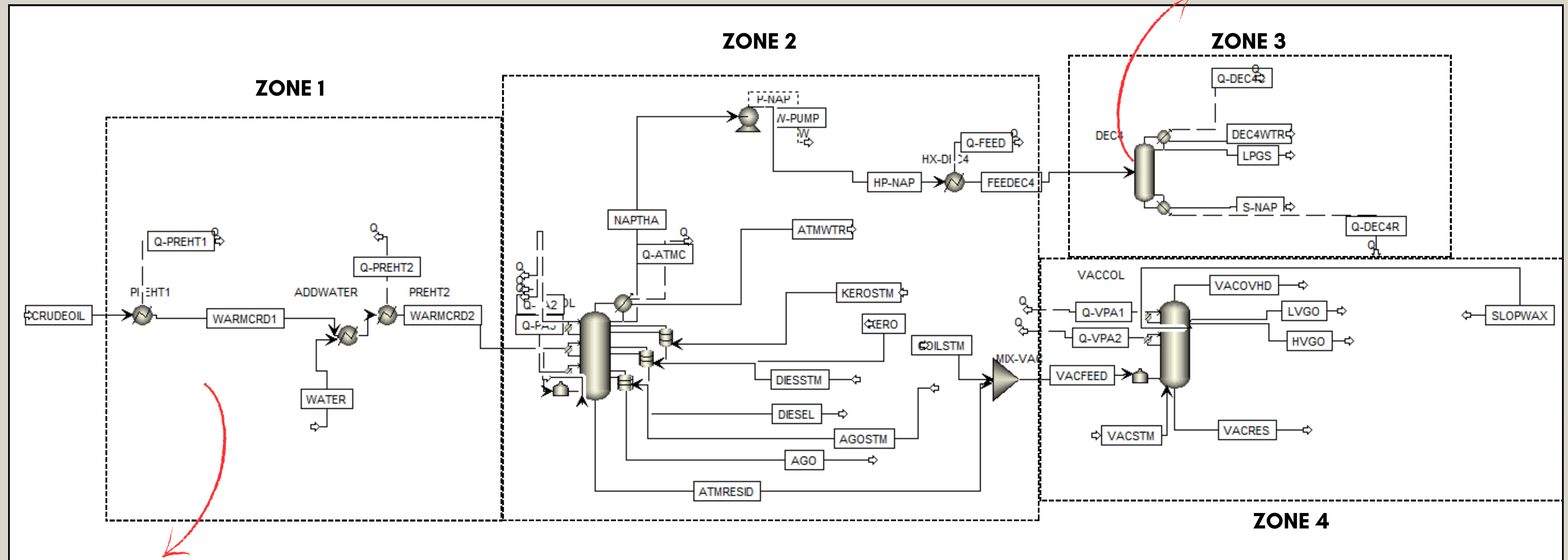


Vacuum Column (VACCOL)

Debutanizer (DEC4)



ASPEN FLOWSHEET



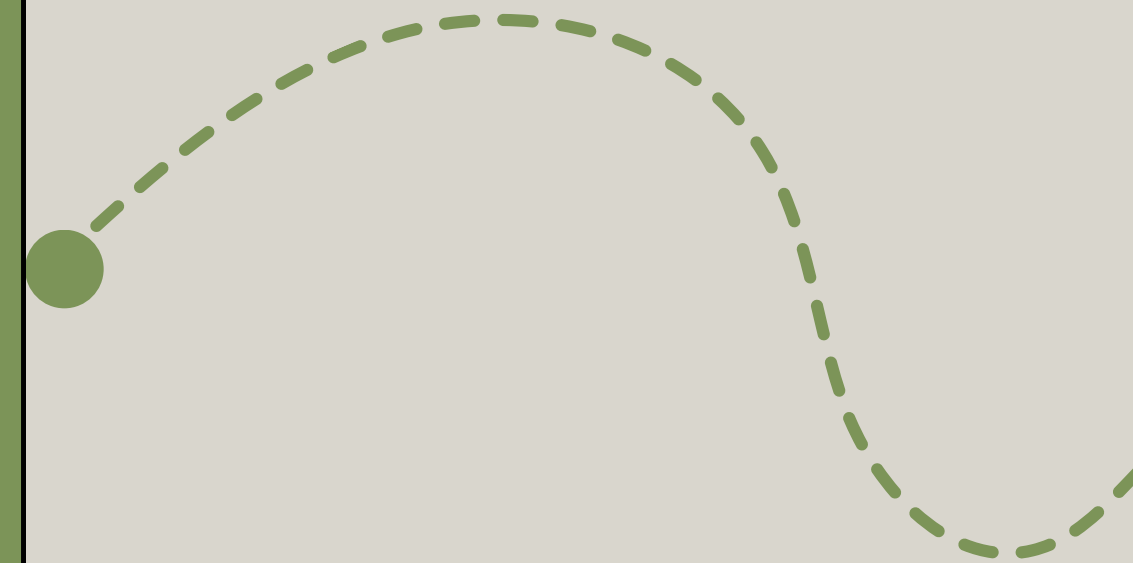
Crude oil Preheat treatment train

Stabilizes overhead naphtha stream

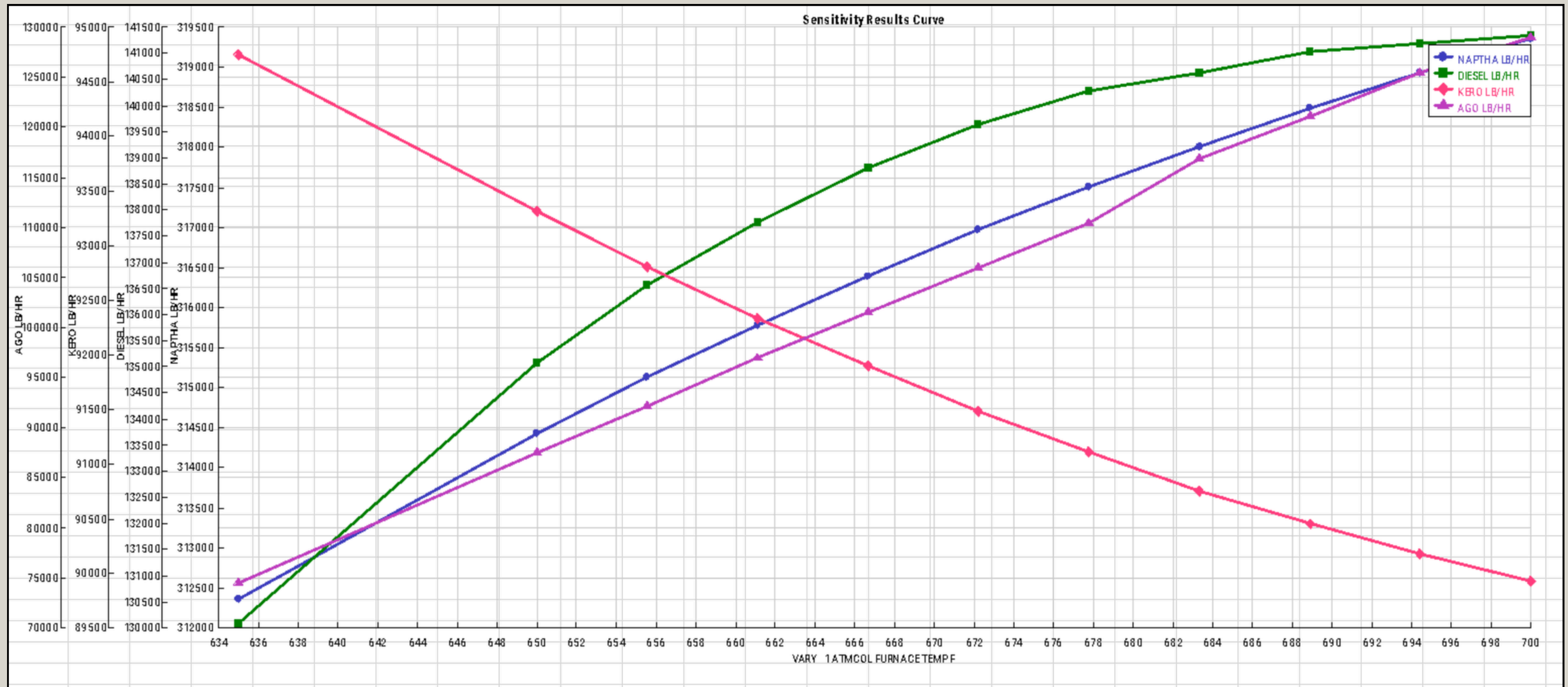
SIMULATION RESULTS: MATERIAL BALANCE

[STREAM TABLE](#)

- **Baseline Feed Rate:** 1,285,641.9 lb/hr
(**CRUDEOIL**).
- Key Product Yields (Baseline):
 - **Naphtha:** 312,350.3 lb/hr.
 - **Diesel:** 312,350.3 lb/hr.
 - **Kerosene:** 94,740.2 lb/hr.
 - **HVGO:** 348,651.9 lb/hr



SENSITIVITY ANALYSIS



- The sensitivity analysis shows that increasing furnace temperature increases naphtha yield, while kerosene decreases due to redistribution of intermediate boiling components. These components shift toward lighter (naphtha) and heavier fractions (diesel and AGO), indicating improved separation and movement of cut boundaries within the column.

SENSITIVITY ANALYSIS: PRODUCT QUALITY

- **Pumparounds:** Total heat removal duty across three PAs (~88 MMBtu/hr). Increasing PA flow improves internal reflux and sharpens product separation.
- **Stripping Steam:** Baseline of 2,500 lb/hr for side products and 20,000 lb/hr for column bottoms. Increasing steam rates enhances the flash point of products by removing light ends

Pinch Analysis

- Method used in process engineering to minimize energy consumption by optimizing heat recovery between hot and cold streams in a system.

Reduce external heating and cooling needs

Improve energy efficiency

Lower operating costs and emissions

Design Guidelines

ABOVE PINCH ($T > 180\text{ }^{\circ}\text{F}$ hot / $160\text{ }^{\circ}\text{F}$ cold)

- Heat sink — receives heat from hot utility (furnace/HP steam)**
- NO cooling utilities allowed above pinch**
- Streams: SLOPWAX, VACRES, ATMRESID, AGO, HVGO, DIESEL, S-NAP, KERO, LVGO**

Pinch

BELOW PINCH ($T < 180\text{ }^{\circ}\text{F}$ hot / $160\text{ }^{\circ}\text{F}$ cold)

- Heat source — rejects heat to cold utility (cooling water / air coolers)**
- NO heating utilities allowed below pinch**
- Streams: LPGS, VACOVHD, HP-NAP, ATMWTR, NAPTHA + cold utility targets**

Design Violation: Transferring heat across the pinch increases utilities above the minimum target!

Hot Stream Data

Stream	Ts (°F)	Tt (°F)	Q (MMBtu/hr)	CP (MMBtu/hr·°F)
SLOPWAX	692.21	100.0	3.19	0.00539
VACRES	691.53	300.0	62.45	0.15950
ATMRESID	612.70	300.0	112.29	0.35910
AGO	560.21	100.0	10.30	0.02238
HVGO	498.70	100.0	33.94	0.08513
DIESEL	489.26	100.0	11.89	0.03055
S-NAP	483.28	100.0	25.73	0.06713
KERO	396.39	100.0	2.75	0.00928
LVGO	383.47	100.0	0.34	0.00120
LPGS	216.84	80.0	34.53	0.25234
VACOVHD	180.00	80.0	228.91	2.28910
HP-NAP	136.22	80.0	66.74	1.18712
ATMWTR	134.81	80.0	233.34	4.25725
NAPTHA	134.81	80.0	67.18	1.22569

Cold Stream Data

Stream	Ts (°F)	Tt (°F)	Q (MMBtu/hr)	CP (MMBtu/hr·°F)
CRUDEOIL	100.0	260.0	184.03	1.15019
Q-PREHT1	100.0	260.0	100.38	0.62738
Q-FEED	136.22	250.0	19.28	0.16945
FEEDC4	136.22	250.0	47.46	0.41712
Q-PREHT2	260.0	450.0	147.30	0.77526
WETCRUDE	260.0	450.0	130.15	0.68500
Q-DEC4R	447.15	483.28	56.21	1.55577
WARMCRD2	450.0	612.70	17.15	0.10541

**Streams to be heated — sorted by supply temperature

Total Cold Duty: 701.96 MMBtu/hr

Heat Balance Check: Hot Duty (893.58) = Cold Duty (701.96) + QH (326.86) – QC (518.48) + correction | Net deficit bridged by utilities

Problem Table Algorithm — Cascade Analysis

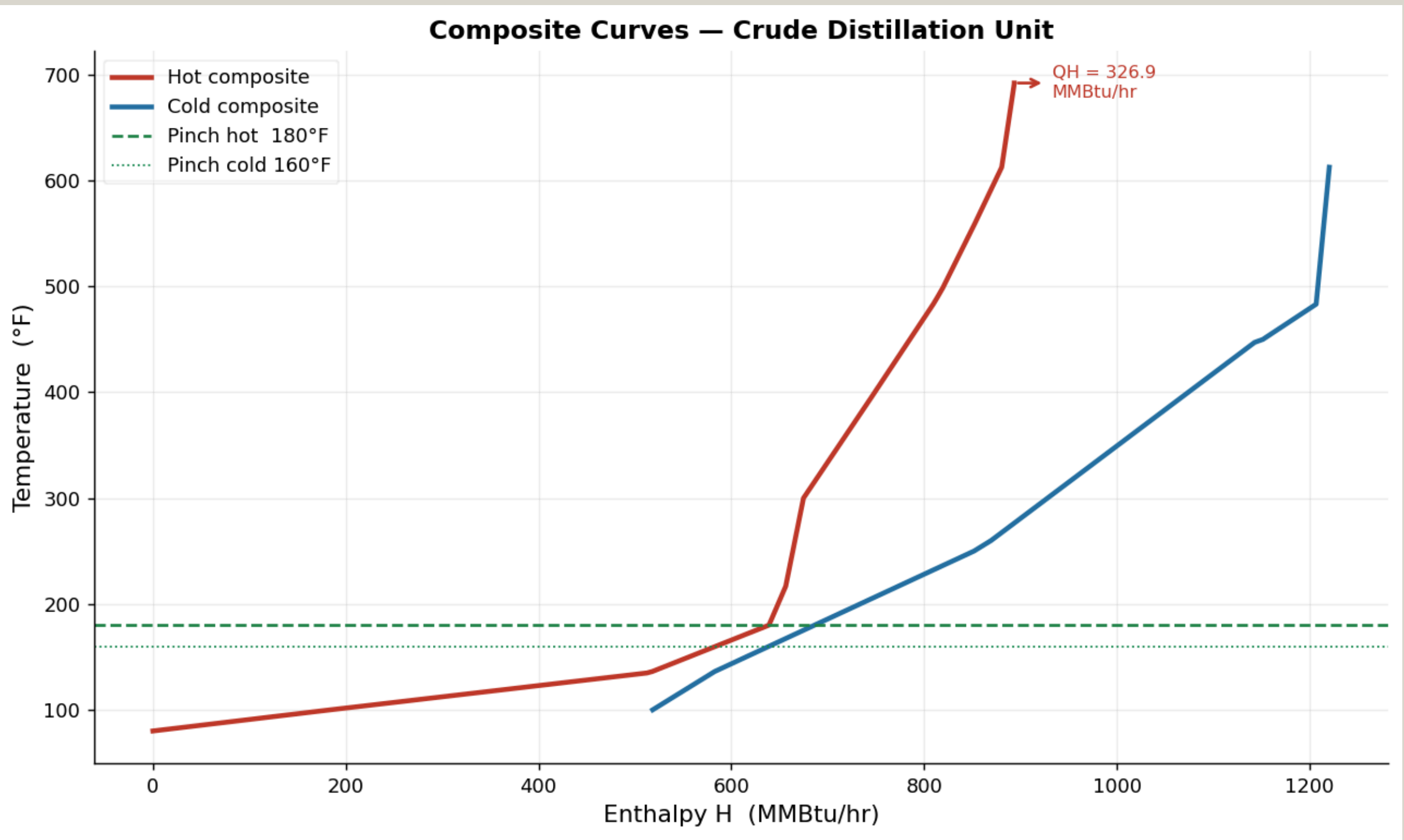
Interval-by-interval heat cascade identifying the pinch

T* High (°F)	T* Low (°F)	ΔT^* (°F)	$\Sigma CP_{\text{hot}} - \Sigma CP_{\text{cold}}$ (MMBtu/hr·°F)	ΔH Interval(MMBtu/hr)
682.21	681.53	0.68	0.0054	0.004
681.53	622.70	58.83	0.1649	9.701
622.70	602.70	20.00	0.0595	1.190
550.21	493.28	56.93	0.4410	25.106
488.70	479.26	9.44	-1.0297	-9.720
479.26	473.28	5.98	-0.9991	-5.975
460.00	457.15	2.85	-2.2869	-6.518
457.15	386.39	70.76	-0.7311	-51.733
373.47	290.00	83.47	-0.7206	-60.148
270.00	260.00	10.00	-1.5565	-15.565
206.84	170.00	36.84	-1.8907	-69.653
170.00	146.22	23.78	0.3984	9.474
146.22	126.22	20.00	0.9849	19.698
124.81	110.00	14.81	7.6550	113.371
90.00	70.00	20.00	9.2115	184.230

★ Pinch at $T^* = 170$ °F (shifted) → Hot side 180 °F, Cold side 160 °F. Minimum cascade heat = 0 at pinch interval. Positive ΔH → heat required. Negative ΔH → heat released

Composite Curves & Pinch Point

Temperature vs. Enthalpy — Hot and Cold composite curves



- Hot Composite Curve: As temperature decreases → heat is released
- Cold Composite Curve: As temperature increases → heat is absorbed

Pinch Hot Side	180 °F
Pinch Cold Side	160 °F
Shifted T*	170 °F
ΔT_{min}	20 °F
QH (min)	326.86 MMBtu/hr
QC (min)	518.48 MMBtu/hr
Max Recovery	566.72 MMBtu/hr

At pinch:

- Heat recovery is maximum
- No extra heat can be transferred

GRAND COMPOSITE CURVE

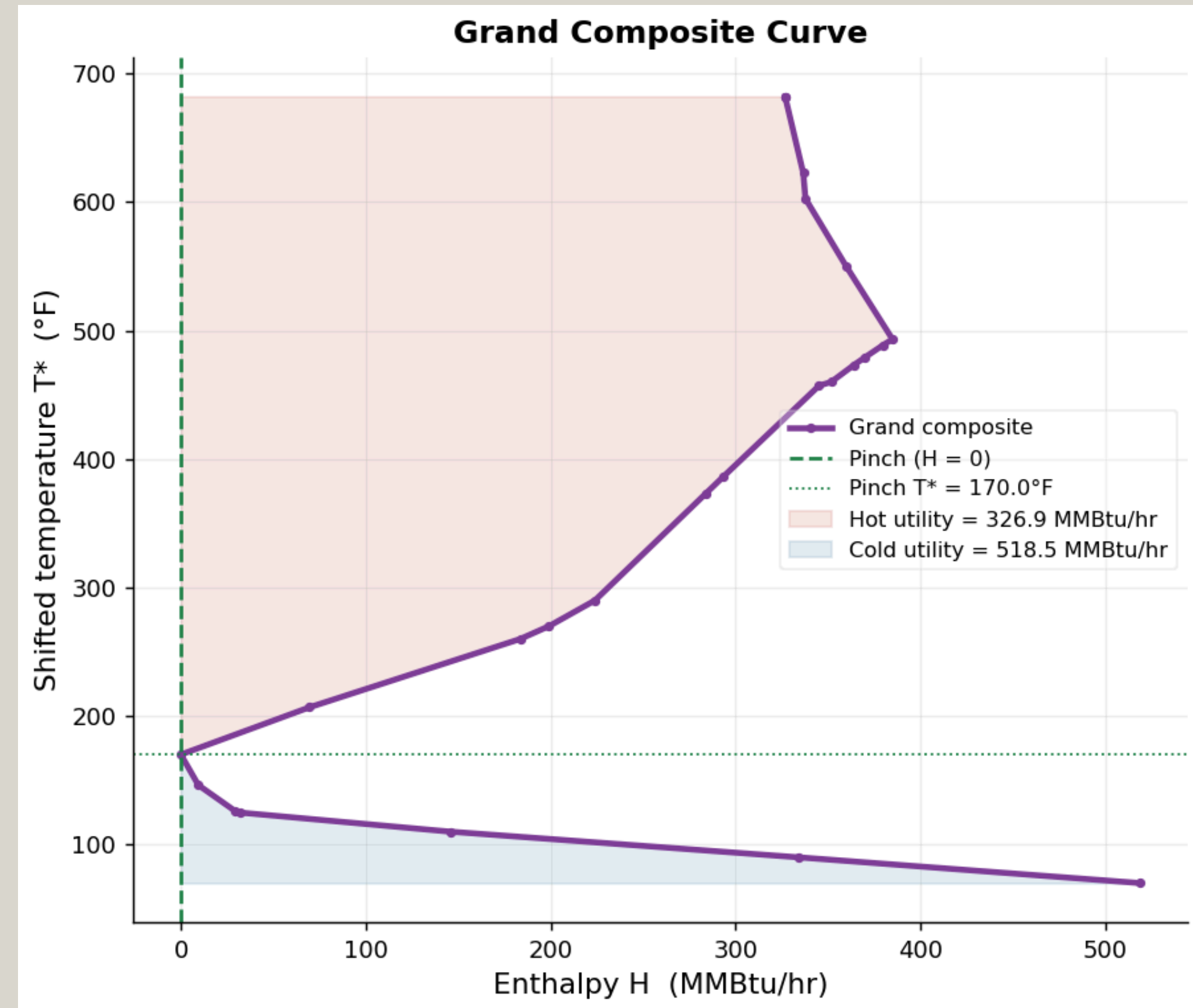
Net heat flow of a process across temperature levels, after combining all hot and cold streams.

Above the pinch (higher temperatures)

- Curve moves to the right
- Indicates excess heat available
- You should recover this heat internally
- Or use it for: Steam generation, Power generation, Other process heating

Below the pinch (lower temperatures)

- Very small region in graph
- Very little heating demand
- Process is heat-rich overall



$\Delta T_{\min}$ Sensitivity Analysis

- Higher $\Delta T_{\min}$ = larger temperature gap required in heat exchangers
- This makes heat recovery less efficient

Result:

- Less heat exchanged internally
- More external utilities needed

Engineering trade-off:

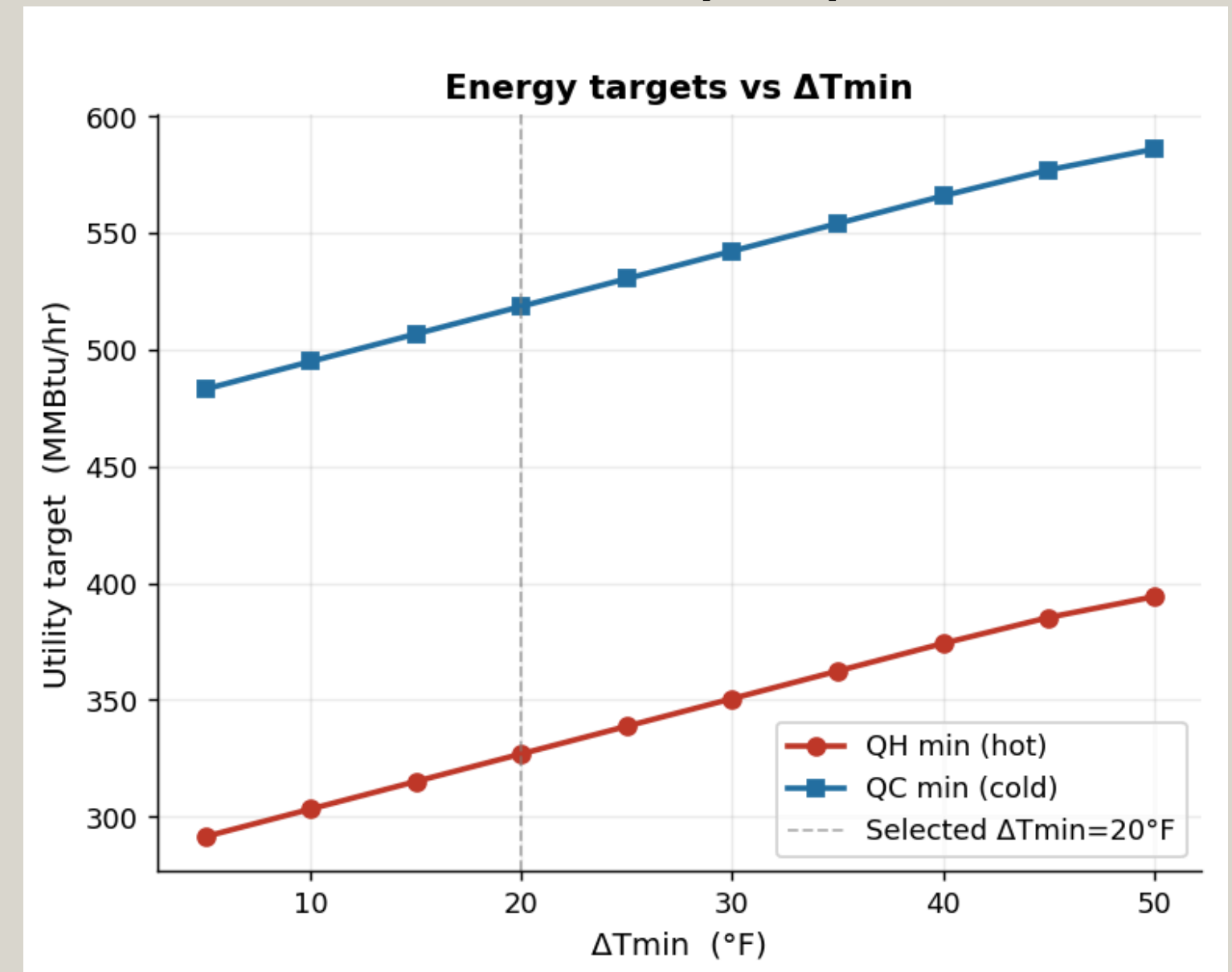
Low $\Delta T_{\min}$ (5–10°F):

- Maximum heat recovery
- Very large heat exchanger area → high capital cost

High $\Delta T_{\min}$ (40–50°F):

- Smaller exchangers → cheaper equipment
- High energy cost (utilities increase)

Del_ T min sensitivity analysis



$\Delta T_{\min}$ increases (5 → 50°F):

QHmin (heating) increases

QCmin (cooling) also increases



THANK YOU